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welche uni angeben, welche reihenfolge..?

Automatic Optimization of Level Parameters for RSHash

(RSHash is a minimizer based fast, compressed hash table for storing DNA kmers. It was recently introduced by the Reinert Group.)

Bachelor's Thesis

by Bjarne David Jehn

to obtain the degree of "Bachelor of Science" (B. Sc.)
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Declaration of Authorship

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Berlin, _____ _____

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Abstract

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1 Introduction

1.1 Motivation

Laut **Nie06** ist unser Thema voll wichtig.

1.2 Stand der Forschung

Mehrere Zitate auf einmal: **Nie06**; **GJ79**;
KN12

2 Notation und Grundlegendes

Die Arbeit sollte so gut wie möglich alle zum Verständnis notwendinge Definition, Notation, Konventionen beinhalten. Dieser Abschnitt ist dafür geeignet.

Graphentheorie. Sei $G = (V, E)$ ein ungerichteter Graph, wobei V die Knotenmenge und $E \subseteq \{\{v, w\} \mid v, w \in V, v \neq w\}$ die Kantenmenge bezeichnen. Für einen Graphen G schreiben wir auch $V(G)$ und $E(G)$ für die Knoten- beziehungsweise Kantenmenge von G.

Wenn man viel Notation einzuführen hat, dann kann ein sogenanntes *labeling* empfehlenswert sein. Wir nennen

$V(G)$	die *Knotenmenge* von G;
$E(G)$	die *Kantenmenge* von G mit $E(G) \subseteq \binom{V(G)}{2}$; die Knoten u und v einer Kante $e = \{u, v\} \in E(G)$ werden *Endpunkte* genannt;
n_G	die Anzahl $\lvert V(G)\rvert$ der Knoten;
m_G	die Anzahl $\lvert E(G)\rvert$ der Kanten.

3 MINLP Modeling

Table 3.1: Constants and Variables for One Level

Constants	
T_{SL}	time to lookup a k-mer in vector S
T_H	time to lookup a k-mer in the hash table
$P_K(m,t)$	empirically estimated value for a k-mer with a minimizer of length m to have a frequency less than t
Variables	
$\mathbf{p_K} \in [0,1]$	probability of a k-mer to have a minimizer with frequency $\leq t$
$\mu_m \in \{0,1\}$	binary selector for every minimizer length $m \in \mathcal{M}$
$\tau_t \in \{0,1\}$	binary selector for every threshold $t \in \mathcal{T}$
$\mathbf{m}$	optimal minimizer length
$\mathbf{t}$	optimal threshold

One Level

As a first approach we model the runtime and space consumption for only one level. Let $\mathcal{M}$ be the Set of all minimizer length and $\mathcal{T}$ the set of all thresholds to be considered.

Runtime

To lookup a k-mer q with minimizer $m(q)$ of frequency $f(m(q))$ on one level, first the bit vector R is checked in order to determin whether $1 \leq f(m) \leq threshold$. This can be ignored when modeling the lookup runtime as this is practically not influenced by the parameters. reason

For $\mathbf{p_K}$ of the k-mers, the corresponding bit in R will be set, which results in a lookup in S. If $m(q)$ does not appear at all or has a $f(m(q)) > t$, a lookup in the hash table will follow. The lookup time T_{lookup} for one level can thus be modeled as:

$$T_{lookup} = T_{SL} * \mathbf{p_K} + T_H * (1 - \mathbf{p_K})$$

In order to determine m and t we introduce binary selector variables μ_m, τ_t for each possible $m \in \mathcal{M}$ and $t \in \mathcal{T}$.
μ_m should be 1 exactly when the optimal minimizer length is m.
τ_t should be 1 exactly when the optimal threshold is t.
The selectors for m and t must be binary and mutually exclusive:

$$\mu_m \in 0,1 \mid \forall m \in \mathcal{M}$$
$$\tau_t \in 0,1 \mid \forall t \in \mathcal{T}$$
$$\sum_{m \in \mathcal{M}} \mu_m = 1$$
$$\sum_{t \in \mathcal{T}} \tau_t = 1$$

For a fixed minimizer length m and threshold t let $P_K(m,t)$ be the empirically determined percentage of k-mers q that have a minimizer with $f(m(q)) < t$.
We can then model $\mathbf{p_K}$ with:

$$\mathbf{p_K} = \sum_{m \in \mathcal{M}} \left(\mu_m \sum_{t \in \mathcal{T}} P_K(m,t) * \tau_t \right)$$

$\mathbf{m}$ and $\mathbf{t}$ can then be determined by:

$$\mathbf{m} = \sum_{m \in \mathcal{M}} \mu_m * m$$
$$\mathbf{t} = \sum_{t \in \mathcal{T}} \tau_t * t$$

The above results in the following MINLP (Mixed Integer Non Linear Program):

MINIMIZE

$$T_{lookup} = {T_{SL}}^1 * \mathbf{p_K} + T_H * (1 - \mathbf{p_K})$$

SUBJECT TO

$$\mathbf{p_K} = \sum_{m \in \mathcal{M}} \left(\mu_m \sum_{t \in \mathcal{T}} P_K(m,t) * \tau_t \right)$$
$$\mu_m \in 0, 1 \mid \forall m \in M$$
$$\tau_t \in 0, 1 \mid \forall t \in T$$
$$\sum_{m \in M} \mu_m = 1$$
$$\sum_{t \in T} \tau_t = 1$$
$$\mathbf{m} = \sum_{m \in \mathcal{M}} \mu_m * m$$
$$\mathbf{t} = \sum_{t \in \mathcal{T}} \tau_t * t$$

For now we will assume that $T_{SL} \approx 40ns$ and $T_H \approx 20ns$.
Solving this MINLP will result in a Solution where the μ_m, τ_t corresponding to the smallest $m \in \mathcal{M}$, $t \in \mathcal{T}$ will be set to one. This is as expected, since this leads to the smallest possible $\mathbf{p_K}$. This minimizes the k-mers found in the vector S and thus maximizes the lookups in the (faster) hash table.

However this also results in a high space consumption, because each entry in the hash table containing the minimizer as a key and the offset as its value is significantly larger than an entry in S which only contains offsets.

In order to take that into account, we have to add a constraint modeling the space consumption.

Space Consumption

The space consumption of the Data Structure used by RSHash can be split into that of the Vectors R, O and S as well as the hash table H.

Bit Vectors R and S

R and S can likely be disregarded when modeling the space consumption. explain why
R is a Bit Vector of length $4^{\mathbf{m}}$ that contains n 1s marking the existence of each encountered minimizer. Using Elias-Fano-Encoding, its size can be compressed to possibly model n?

$$n\lceil log(4^{\mathbf{m}}/n) \rceil + 3n$$

Offsetvector O

O contains one entry of size $log|T|$ for the offset of every minimizer with frequency $\leq t$ in T.
Every minimizer of length $\mathbf{m}$ covers an average of $\frac{k-\mathbf{m}+2}{2}$ consecutive k-mers. We can thus model the bits per k-mer in O as:

$$bpk_O = \mathbf{p_K} * \frac{2}{k - \mathbf{m} + 2} * log|T|$$

Hash table H

For $1 - \mathbf{p_K}$ of the k-mers q in T H contains an entry $(key : q, value : offset(q))$
Each entry has size $size(key) + size(value) = 64 + log|T|$.

Table 3.2: Constants and Variables for Several Level

Constants	
T_{SL}	time to lookup a k-mer in vector S
T_H	time to lookup a k-mer in the hash table
$P_K(m,t)$	empirically estimated value for a k-mer with a minimizer of length m to have a frequency less than t
Variables	
$\mathbf{p_K^l} \in [0,1]$	probability of a k-mer to have a minimizer with frequency $\leq t$ on level l
$\mu_m^l \in \{0,1\}$	binary selector for every minimizer length $m \in \mathcal{M}$ on level l
$\tau_t^l \in \{0,1\}$	binary selector for every threshold $t \in \mathcal{T}$ on level l
$\mathbf{m}^l$	optimal minimizer length on level l
$\mathbf{t}^l$	optimal threshold on level l

The bits per k-mer required for the hash table can thus be modeled by

$$bpk_H = (1 - \mathbf{p_K}) * (64 + log|T|)$$

Several Levels

Let

Runtime

RSHash uses several levels, to make use of the skewniss of minimizer distribution.
Each level l has a pair of parameters $\mathbf{m}^l, \mathbf{t}^l$ which result in a probability $\mathbf{p_K^l}$ of a k-mer belonging to level l.
For now, we still assume, that the time for a lookup in the vectors R, S as well as the hash table H are (practically) independent of the parameters $\mathbf{m}^l, \mathbf{t}^l$.

reason

When using L levels, the probability of a k-mer belonging to a higher level is dependent on the probability of k-mers belonging to the levels below.
On each level $1 \leq l \leq L$, for $\mathbf{p_K^l}$ of the k-mers, the lookup will be completed by finding a value in the offset vector on that level whilst the remaining $(1 - \mathbf{p_K^l})$ queries result in a lookup on the next level.
We can model the lookup time for all levels $1 \leq l < L$ levels recursively as:

$$T_{lookup}^l = \begin{cases} \mathbf{p_K^l} * T_{SL} + (1 - \mathbf{p_K^l}) * T_{lookup}^{l+1} & | \, 1 \leq l \leq L \\ T_H & | \, l = L + 1 \end{cases}$$

For $L = 3$ levels, this results in:

$$\begin{aligned} T_{lookup} = \; & \mathbf{p_K^1} * T_{SL} \\ & + (1 - \mathbf{p_K^1}) * \Big(\mathbf{p_K^2} * T_{SL} \\ & \qquad + (1 - \mathbf{p_K^2}) * \Big(+ \mathbf{p_K^3} * T_{SL} \\ & \qquad\qquad + (1 - \mathbf{p_K^3}) * \end{aligned}$$

Because $\mathbf{m}^l$ and $\mathbf{t}^l$ can be different on every level l, a different set of binary selector variables μ_m^l and τ^l is needed.
The constraints for each level are the same as for one level:

$$\begin{aligned} & \mu_m^l \in 0,1 && | \, \forall (l,m) \in (\mathcal{L} \times \mathcal{M}) \\ & \tau_t^l \in 0,1 && | \, \forall (l,t) \in (\mathcal{L} \times \mathbf{t}) \\ & \textstyle\sum_{m \in \mathcal{M}} \mu_m^l = 1 && | \, \forall l \\ & \textstyle\sum_{t \in \mathcal{T}} \tau_t^l = 1 && | \, \forall l \end{aligned}$$

For every level l $\mathbf{p_K^l}, \mathbf{m}^l$ and $\mathbf{t}^l$ can also be modeled analogously to having only one level:

this only holds under the assumption, that the mini-

$$\mathbf{p}_\mathbf{K}^l = \sum_{m \in \mathcal{M}} \left(\mu_m^l \sum_{t \in \mathcal{T}} P_K(m,t) * \tau_t^l \right) \quad | \; \forall l \in \mathcal{L}$$

$$\mathbf{m}^l = \sum_{m \in \mathcal{M}} \mu_m^l * m \quad | \; \forall l \in \mathcal{L}$$

$$\mathbf{t}^l = \sum_{t \in \mathcal{T}} \tau_t^l * t \quad | \; \forall l \in \mathcal{L}$$

paragraph/section about the complexity of finding the solution

For these reasons, an approach yielding a solution that is a certain approximation of the optimum but takes less time to find it could be of interest.

An easy adaption to speed up the search for empirical values would be to only sample part of the genome. By sampling sequences of length at a large number of random locations, the probability of choosing a bad representation becomes highly unlikely.

what length

how lorge

By sampling only part of the genome, we can speed up the finding of empirical values by a factor of / or even "get it down to a constant" (that is for a fixed number of sample locations - regardless of the genomes size. this however results in a growing error for larger genomes).

factor

Space Consumption

Offsetvector O

should I introduce an index for the minimizer function as it differs for every level? e.g. $m^l(q)$

For several levels, the offset vector for level l only contains an entry for a k-mer q, if the k-mers minimizer $m(q)$ has a frequency $f(m(q)) > t^i$ for all levels $1 \leq i < l$ and $f(m(q)) \leq t^l$.

The bits per k-mer for the Offset vector on level l can thus be modeled as:

$$bpk_{O^l} = \prod_{i=1}^{l-1} (1 - \mathbf{p}_\mathbf{K}^i) * \mathbf{p}_\mathbf{K}^l * \frac{2}{k - \mathbf{m}^l + 2} * log|T|$$

Hash table H

The hash table only contains an entry k-mers q, for which $f(m(q)) > t^l$ for all levels $1 \leq l \leq L$.

The bits per k-mer for the hash table can thus be modeled as

$$bpk_H = \prod_{l=1}^{L} (1 - \mathbf{p}_\mathbf{K}^l) * (64 + log|T|)$$

Limitations

over what

does it?

how much, why?

This approach of solving the constructed model has the advantage () of finding optimal results .

However for large genomes, finding the empirical values $P_K(m,t)$ becomes increasingly time consuming and needs a lot of RAM .

4 Subsampling

Motivation

The time for processing a genome in order to find the minimizer distribution $\mathbf{p_K}(m,t)$ for possible minimizer lengths $m \in \mathcal{M}$ and thresholds $t \in \mathcal{T}$ is directly proportional to the size of the genome.
For large genomes, this processing can become the dominating factor when searching for optimal parameters using the MINLP Modeling approach. In order to speed up the search for the empirical values, we introduce a subsampling method.

Subsampling Method

Instead of calculating the distribution over the full genome, we sample n sequences of length $l \geq k$ at random positions in the genome.

how to handle overlapping samples

Time Complexity of Finding the Minimizer Distributions

genaue zeitliche abhängigkeit 1. verstehen und 2. aufschreiben

Time Complexity using Subsampling

By applying the proposed subsampling strategy, the original genome T of length $|T|$ is reduced to the subsampled text T^S of length $|T^S| = n * l$. This means a reduction in size by a factor of $\frac{|T|}{|T^S|}$.
Depending on the choice of n and l this results in a speedup of the processing time by a factor of or even a constant processing time when both parameters are chosen as a constand regardless of $|T|$ (this however should result in a growing error for larger genomes).

assuming no overlaps

factor

time

Error Analysis

Choosing l

The length l has to be at least k.

wenn $l > k$ ist, überlappen sich mehrere kmere an den gesampleten stellen. ich verstehe noch nicht, warum genau dass von vorteil ist.
Ist es für die Konstruktion der Burrow-Wheeler-Transformation relevant, bze. findet diese hier überhaupt statt?
oder lässt sich ein streaming case für ministats implementieren/ ist dies bereits so umgesetzt?

Choosing n

Choosing a constant n would result in a processing time that would no longer be dependent of the genomes length $|T|$, as the sampled text would have constant size $n * l$. For large genomes, the percentage $\frac{|T^S|}{|T|}$ of the genome represented in the subsample can become arbitrarily small, resulting in growing error.

assuming no overlaps

If instead n is chosen relative to $|T|$, $\frac{|T^S|}{|T|}$ will have a constant value independent of the genomes size. Even though this means, that the time needed, in order to find the minimizer distributions would still be proportional to the genomes size, this should be the preferred option as it results in a bounded error of the approximation.

For a large enough n, the probability of choosing a bad representation becomes highly unlikely.

zu vage, begründen

Parametrisierte Komplexität. Wir verwenden Standardnotation aus der Parametrisierten Komplexität und Algorithmik [**Nie06**].

5 LaTeX-Beispiele

5.1 Grundlegendes

Einzelne Zeilenumbrüche sind Latex egal. Mit einem doppelten Umbruch

erzeugt man einen neuen Absatz. Der Übersicht halber sollte man die Zeilen im Latex-Quellcode trotzdem nicht zu lang machen.

Mit newcommand lassen sich eigene Befehle definieren: Jede supercalifragilisticexpialigetische Instanz lässt sich mit einem supercalifragilisticexpialigetischen Algorithmis in supercalifragilisticexpialigetischer Zeit lösen. (Sehr nützlich, wenn man später Dinge umbenennen will.)

5.2 Formatierung

Ich bin kursiv. **Ich bin fett**. Ich bin in Kapitälchen (für Problemnamen). "Ich stehe in Anführungszeichen."

Hier kommt eine Formel: $a = b + c$. Die nächste Formel kriegt mehr Platz und wird am &-Zeichen ausgerichtet:[1]

$$\begin{aligned} f : \mathbb{R} &\to \mathbb{R} \\ x &\mapsto x^2 \end{aligned}$$

Ohne Sternchen gibt's Zeilennummern:

$$a = |x - 2| \tag{5.1}$$
$$b = \|a + x\|_2 \tag{5.2}$$
$$c = a^2 + b^2 \tag{5.3}$$
$$\sqrt{17} > 1 \tag{5.4}$$

[1] Man kann auch an mehreren Stellen ausrichten.

Hoch- und tiefgestellt wird so: $x_1, x^2, x_{i,j}^t$

Brüche schreibt man $\frac{n!}{k!(n-k)!}$ oder im Fließtext auch einfach $n!/(k!(n-k)!)$.

Ein paar weitere Operatoren:

$$\sum_{i=1}^{n} \int_a^b \lim_{x \to \infty} \max_{y \geq x} \left\{ x \cdot y \middle| \left[x^2 - y^2\right] \subseteq \tilde{\Omega} \right\}$$

Nützliche Buchstaben: $\ell, \theta, \vartheta, \epsilon, \varepsilon, \phi, \varphi$

Nützliche Relationen: $G \neq H \simeq K \cong T \approx \emptyset$.

Wer den Namen für ein bestimmtes Symbol sucht schaut am besten auf http://detexify.kirelabs.org/

Wenn man Text innerhalb einer Formel einfügen möchte, muss dieser mit $\backslash text$ ausgezeichnet werden:

$V = \frac{4}{3}\pi z^3$ Volumen einer Kugel mit

$V = \pi z^2 a = \pi z z a$ Volumen einer Pizza mit Radius z un

Versuche Zeilenumbrüche an mathematischen Ausdrücken wie den nun hier folgenden G zu vermeiden. Verwende die Tilde im Latex-Code um das zu vermeiden, wie folgt.

Versuche Zeilenumbrüche an mathematischen Ausdrücken wie den nun hier folgenden G zu vermeiden.

Und noch was: Sätze sollten auch nicht mit mathematischen Ausdrücken beginnen. Also, statt "G ist toll." besser "Graph G ist toll.".

5.3 Theoreme & Co

Definition 5.1. Für $a, b \in \mathbb{R}$ schreiben wir $a < b$ falls gilt

hier feht noch was. Manchmal ist es sinnvoll etwas lokal statt in Kapitel 2 zu definieren.

Lemma 5.2 (trivial)**.** $a < b \iff b > a$

Proof. Gilt nach Definition 5.1. □

Theorem 5.3 (Robertson, Seymour)**.** *Die Menge der endlichen Graphen ist wohlquasigeordnet bzgl. der Relation "$\preceq$".*

Proof. Folgt aus Lemma 5.2 in Verbindung mit Gleichung (5.4) (Übungsaufgabe). □

Vertex Cover
Eingabe: Ein Graph G, $k \in \mathbb{N}$.
Frage: $\exists C \subseteq V(G) : |C| = k$ und $\forall e \in E(G) : e \cap C \neq \emptyset$?

Beobachtung 5.4. Vertex Cover *ist lösbar in exponentieller Zeit.*

Aus **??** 5.4 folgt unmittelbar das folgende Korollar 5.5.

Korollar 5.5. Vertex Cover *ist entscheidbar.*

Example 1. Ein interessantes Beispiel.

Remark. Eine Anmerkung.

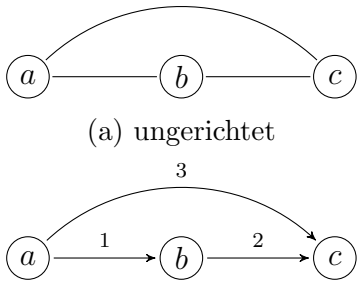
5.4 Abbildungen

Beispiele für Graphen sind in Abbildung 5.1 abgebildet. Hierbei ist Abbildung 5.1b ein *gerichteter Graph* (auch *Digraph*).

Wer zu faul ist, die Bilder in tikz zu erstellen, kann sich auch `http://ipe.otfried.org/` anschauen.

Tabelle 5.1 ist eine große Tabelle. Normalerweise versucht Latex selbst, einen geeigneten Ort für Abbildungen zu finden.

Externe Bilder lassen sich auch einbinden (siehe Abbildung 5.2).



(a) ungerichtet

(b) gerichtet

Figure 5.1: Beispiele für Graphen. Achtung: *label* immer **nach** *caption*, sonst gehen die Verweise kaputt.

Table 5.1: Ergebnisse der Laufzeitexperimente. Alle Zeiten sind in Sekunden angegeben. Im Gegensatz zu Abbildungen, Tabellenüberschriften sind über der Tabelle zu platzieren.

	file	n	m	WCC [s]	HKN [s]	CHI
Clustering	email	1K	5K	0.67	48.75	
	hep-th	7K	15K	0.43	1.35	
	netscience	1K	2K	0.24	0.37	
	PGPgiantcompo	10K	24K	0.67	1.70	
Co-author	citationCiteseer	268K	1.1M	21.46	83.90	
	coAuthorsCiteseer	227K	0.8M	11.46	50.35	
	coAuthorsDBLP	289K	0.9M	15.76	78.51	
	coPapersCiteseer	434K	16M	—	—	

5.5 Pseudocode

Algorithmus 1 ist Euklids Algorithmus zur Bestimmung des größten gemeinsamen Teilers zweier Zahlen. Die Schleife in Algorithmus 1 terminiert sobald a durch b teilbar ist.

Für mehr Informationen zu Pseudocode in LaTeX, empfiehlt sich ein Blick in die algorithmicx-Anleitung [**algorithmicx**].

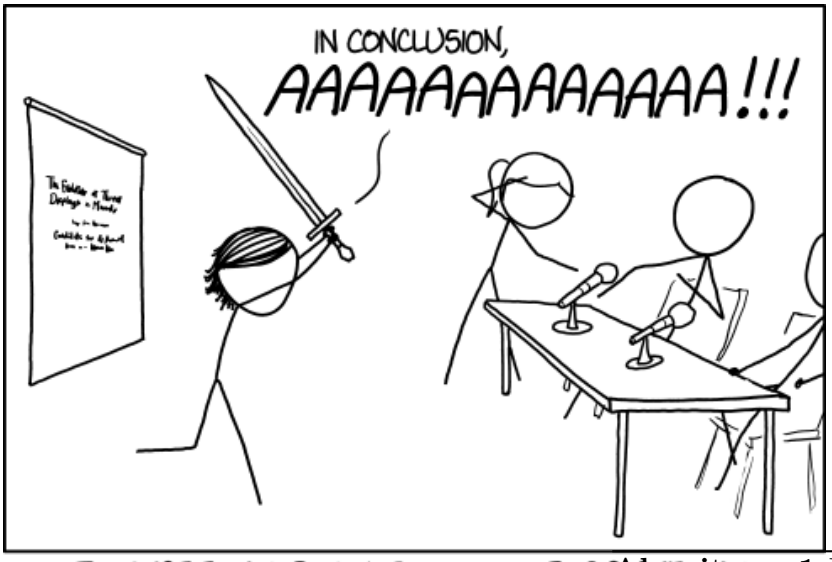


Figure 5.2: MY RESULTS ARE A SIGNIFICANT IMPROVEMENT ON THE STATE OF THE AAAAAAAAAAAART [**xkcd1403**]

Algorithmus 1 Euklids Algorithmus

Input: $a, b \in \mathbb{N}$
Output: Der größte gemeinsame Teiler von a, b

1: **function** EUKLID(a, b)
2: $r \leftarrow a \bmod b$ ▷ r ist der Rest der Ganzzahl-Division a/b
3: **while** $r \neq 0$ **do**
4: $a \leftarrow b$
5: $b \leftarrow r$
6: $r \leftarrow a \bmod b$
7: **end while**
8: **return** b
9: **end function**

Table 5.2: Bonustabelle (automatisch generiert aus example-data.csv) Die Beschriftung ist immer über Tabellen aber unter Bildern...

Graph	n	m	blossom5	kernel	n'	m'	blossom5'
as20000102	6,474	12,572	0.03	0.01	6	9	0.01
email-Eu-core	1,005	16,064	0.04	0.01	869	12,945	0.03
ca-GrQc	5,242	14,484	0.02	0.02	2,129.1	8,110.2	0.02
p2p-Gnutella08	6,301	20,777	0.07	0.01	2,259	7,020	0.02
ca-HepTh	9,877	25,973	0.04	0.02	4,592.8	13,042.6	0.02
p2p-Gnutella09	8,114	26,013	0.08	0.01	2,404.4	7,297.4	0.02
p2p-Gnutella06	8,717	31,525	0.1	0.02	5,205	17,443	0.03
p2p-Gnutella05	8,846	31,839	0.1	0.02	5,265	17,674	0.03
p2p-Gnutella04	10,876	39,994	0.13	0.02	6,288	20,618	0.03
p2p-Gnutella25	22,687	54,705	0.09	0.02	142.3	245.8	0.01
facebook-combined	4,039	88,234	0.12	0.03	3,807	83,668	0.09
ca-CondMat	23,133	93,439	0.11	0.05	16,681.7	71,710.1	0.07
ca-HepPh	12,008	$1.2 \cdot 10^8$	0.16	0.05	8,287.4	91,214.3	0.08
email-Enron	36,692	$1.8 \cdot 10^8$	0.27	0.05	16,554	68,656.2	0.06

6 Conclusion

Ein Fazit is keine Zusammenfassung. Grob sollten hier erzielte Ergebnisse diskutiert und ein Ausblick für weitere Forschung gegeben werden.

Wir habe dieses tolle Ergebnis. Allerdings blieb diese andere tolle Frage offen. Wir vermuten, dass die Frage positive zu beantworten ist wegen diesem tollen Argument.